POAMS: LECTURE 13
ORBITAL MOTION

I now turn my attention to the phenomenon of true physical motion, which, according to Newton, is everywhere under the influence of the 'universal force of gravity', and hence never truly rectilinear, as studied in Special Relativity. In this way, I will extend our Normal Realist agenda to include the study of orbital motion. I begin this study in this lecture by examining Newton's theory of gravitation and the central role of angular momentum. This sets the scene for the Pope-Osborne Angular Momentum Synthesis (POAMS) introduced in the subsequent lecture. The current lecture is based on some of Chapter 6 of 'Light-speed....'.

1. Natural orbital motion

Our study of general particle motion in these lectures began with *uniform* relative motion. This is for no other reason than that uniform motion is the simplest type of motion to study, certainly as far as the mathematics is concerned. However, in complete contrast to Newton, we do not believe that all free motion is naturally uniform and so rectilinear but rather than that it is naturally curved and ultimately *orbital* relative to some reference point. In this lecture, we begin to investigate the agency whereby the motions of the planets and stars are naturally orbital, the agency which Newton explained in terms of a 'gravitational force'. However, from our Normal Realist point of view, for the same reason that Einstein's concept of 'light travelling *in vacuo*' is unempirical, so is Newton's invisible *in vacuo* 'force of gravity'.

For initial simplicity, we shall assume that the speeds of any moving objects considered are very much less than c, so that any relativistic effects can be ignored. The relative effects of faster motions will be studied in later lectures. For now, I will compare and contrast two very distinct ways in which the phenomenon of orbital motion can be explained. The first way is in the classical terms of a balance between centrifugal force and the centripetal 'force of gravity'. The second way is in Normal Realist terms of *pure angular momentum*. As I will demonstrate, this latter way does not require the classical concept of equally opposing and therefore cancelling invisible forces of 'gravitational attraction' and 'centrifugal acceleration'.

Before I present our Normal Realist approach to orbital motion, beginning in the next lecture, it will prove very useful for me to present the central ideas and results from Newton's theory.

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2. Newton's theory of gravitation

Before the time of Isaac Newton, the observed fact that objects that were dropped always fall towards the ground could not be explained. Another mystery was the strange influence the Sun seemed to exert on the planets to keep them in their apparently closed orbits. Johannes Kepler (1571-1630), a German mathematician and astronomer, had postulated, on the basis of his observations and recordings, that the orbits followed by the planets were ellipses, but he could not explain why. Newton addressed these questions by adding to his three laws of motion outlined briefly at the start of Lecture 9, another law called his *Law of Gravitation* in which he postulated the existence of a universal *in vacuo* force of attraction. This was formulated as follows:

Every object attracts every other object with a **gravitational force, f,** *with magnitude*

$$f = \|\mathbf{f}\| = \frac{GmM}{r^2},$$

where m and M are the masses of the objects concerned, r is the distance between them and G is a constant, known as the **gravitational constant.**

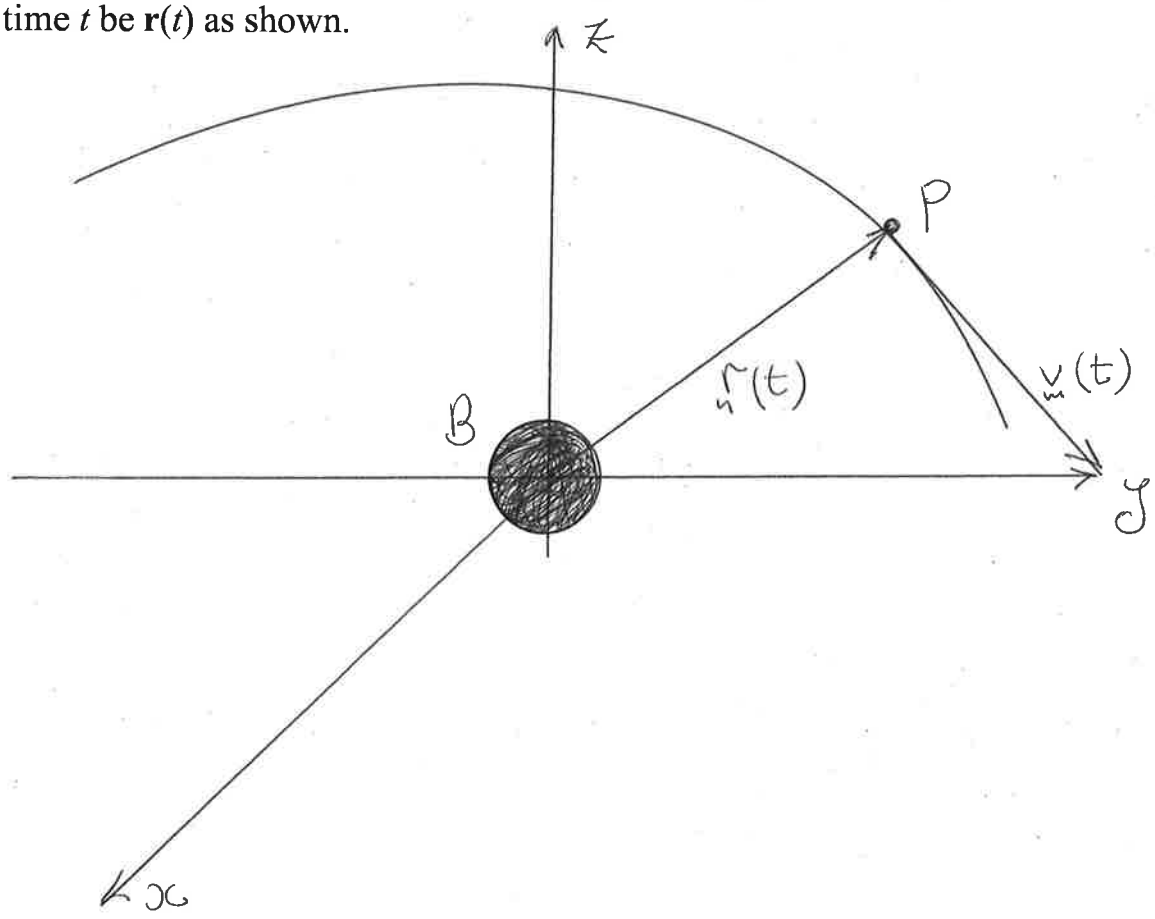
The view underlying this assumption was that of the observable universe as a self-sufficient three-dimensional Euclidean space, with time acting as a *universal* parameter (*i.e.*, not observer dependent), together with an invisible *in vacuo* gravitational force which distorted the (assumed) natural straight-line paths of freely moving objects to form orbital trajectories.

I shall now discuss the classical problem of how the motions of a pair of bodies orbiting each other in an isolated gravitational system are to be calculated. This so-called 'two-body problem' is the only problem concerning orbital motion that is mathematically soluble (without using

numerical programs); any more than just two bodies pose a notoriously intractable problem.

The two-body problem

Let us consider, then, an isolated system consisting of a particle P of mass m under the influence of a body B of mass M, with m very much less than M, as in the case, for example, of the Moon orbiting Earth or a planet orbiting the Sun. (More generally, it is possible to consider a case in which the two masses are more comparable and, in this case, we study the orbits of both bodies around their common *centre-of-mass*.) Let the centre-of-mass of the body B lie at the origin, O, of a set of Cartesian coordinates and the position vector of the particle P, in relation to O, at any particular time t be $\mathbf{r}(t)$ as shown.



According to Newtonian theory, the particle P is attracted towards the body B by a gravitational force, $\mathbf{f}$, of magnitude GmM/r^2, where $r = \|\mathbf{r}\|$, so that by *Newton's Second Law*, [*Lecture 9, equation (2a)*] the acceleration, $\mathbf{a}(t)$, of P, at any instant of time t, is

$$\mathbf{a}(t) = \frac{\mathrm{d}^2\mathbf{r}}{\mathrm{d}t^2} = -\frac{GM}{r^2}\mathbf{n} \quad (1)$$

where **n** is a vector of unit length in the direction of **r**. The differential equation (1) provides an *equation of motion*, since it determines the possible paths of P relative to B. Note that from equation (1), **a** points in the opposite direction to **r**. Here, the gravitational force is a particular example of a *central force*. Such a force always acts in the direction opposite to that of the position vector.

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3. The role of angular momentum in orbital motion

The concept of angular momentum plays a central role in the study of orbital motion. In the two-body problem just described, the **angular momentum, L**, of the particle P relative to the chosen origin, O, is defined by

$$\mathbf{L} = \mathbf{r} \times m\mathbf{v} = \mathbf{r} \times \mathbf{p} \quad (2)$$

where $\mathbf{v}(t) = \mathrm{d}\mathbf{r}/\mathrm{d}t$ is the velocity of P, which always lies tangentially to the path of P, as indicated in the figure above, $\mathbf{p}(t)$ is the **linear momentum** of P and '×' denotes the *vector (cross) product*. It follows that the angular momentum of P is a vector that lies perpendicular to the plane containing **r** and **v**, and so gives information about the orientation of the path of P relative to the origin. Its magnitude is given by

$$L = \|\mathbf{L}\| = mvr \sin\phi$$

where $v = \|\mathbf{v}\|$ is the speed of P and ϕ is the angle between **r** and **v**, which is, in general, is *not constant.*

Now in general, it follows by definition (2) and Newton's Second Law that

$$\frac{\mathrm{d}\mathbf{L}}{\mathrm{d}t} = \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t} \times m\mathbf{v} + \mathbf{r} \times m\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = \mathbf{v} \times m\mathbf{v} + \mathbf{r} \times m\mathbf{a} = \mathbf{r} \times \mathbf{f} \qquad (3)$$

where **f** is the force exerted on *P*, since $\mathbf{v} \times \mathbf{v} = \mathbf{0}$. Here, $\mathbf{r} \times \mathbf{f}$ is the *torque* acting on the particle *P*. In Newtonian theory, the gravitational force **f** is parallel to **r** in the two-body problem so that $\mathrm{d}\mathbf{L}/\mathrm{d}t = \mathbf{0}$ and hence **L** is a *constant* vector. It then follows that the trajectory of *P* relative to the body *B* lies in a plane and that the magnitude, *L*, of the angular momentum is constant.

Since the orbit of the particle *P* lies in a plane, it is convenient to describe it using *plane polar coordinates r* and θ, where $r = \|\mathbf{r}\|$ as before and θ is the angle between the radial vector **r** and some fixed radial axis. The **angular speed** of *P* is then defined as $\mathrm{d}\theta/\mathrm{d}t$ and is denoted by ω. Relative to plane polar coordinates, the position vector, **r**, of *P* is given by

$$\mathbf{r} = (x, y, z) = (r\cos\theta,\ r\sin\theta, 0) = r\mathbf{n} \qquad (4a)$$

where $\mathbf{n} = (\cos\theta, \sin\theta, 0)$, and so is a vector of *unit length* in the direction of **r**. It follows that *P*'s velocity, **v**(*t*), is given by

$$\mathbf{v} = \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t} = \frac{\mathrm{d}r}{\mathrm{d}t}\mathbf{n} + r(-\sin\theta, \cos\theta, 0)\frac{\mathrm{d}\theta}{\mathrm{d}t} = \frac{\mathrm{d}r}{\mathrm{d}t}\mathbf{n} + r\omega\mathbf{m} \qquad (4b)$$

where $\mathbf{m} = (-\sin\theta, \cos\theta, 0)$, another vector of *unit length*. The (orbital) speed, *v*(*t*), of the particle *P* relative to *O* is then given by

$$v^2 = \mathbf{v}.\mathbf{v} = \left(\frac{\mathrm{d}r}{\mathrm{d}t}\right)^2 \mathbf{n}.\mathbf{n} + 2r\omega\frac{\mathrm{d}r}{\mathrm{d}t}\mathbf{n}.\mathbf{m} + r^2\omega^2\mathbf{m}.\mathbf{m} = \left(\frac{\mathrm{d}r}{\mathrm{d}t}\right)^2 + r^2\omega^2 \qquad (5)$$

noting that **n**.**m** = 0. Equation (4b) also tells us that the acceleration, $\mathbf{a}(t) = \mathrm{d}\mathbf{v}/\mathrm{d}t$, of *P* is given by

$$\mathbf{a} = \frac{\mathrm{d}^2 r}{\mathrm{d}t^2}\mathbf{n} + \frac{\mathrm{d}r}{\mathrm{d}t}(-\sin\theta, \cos\theta, 0)\frac{\mathrm{d}\theta}{\mathrm{d}t} + \left(\frac{\mathrm{d}r}{\mathrm{d}t}\omega + r\frac{\mathrm{d}\omega}{\mathrm{d}t}\right)\mathbf{m} + r\omega(-\cos\theta, -\sin\theta, 0)\frac{\mathrm{d}\theta}{\mathrm{d}t}$$

$$\Rightarrow \mathbf{a} = \left(\frac{\mathrm{d}^2 r}{\mathrm{d}t^2} - r\omega^2\right)\mathbf{n} + \left(2\frac{\mathrm{d}r}{\mathrm{d}t}\omega + r\frac{\mathrm{d}\omega}{\mathrm{d}t}\right)\mathbf{m} \quad (6)$$

Finally, using definition (2) together with equations (4a) and (4b), the angular momentum of P, in terms of polar coordinates, is given by

$$\mathbf{L} = r\mathbf{n} \times m\left(\frac{\mathrm{d}r}{\mathrm{d}t}\mathbf{n} + r\omega\mathbf{m}\right) = mr^2\omega(\mathbf{n} \times \mathbf{m}) = mr^2\omega(0, 0, 1)$$

so that its magnitude is

$$L = mr^2\omega \quad (7)$$

Since L is a constant, it follows by equation (7) that $r^2\omega$ is a constant, so that differentiating gives

$$2r\omega\frac{\mathrm{d}r}{\mathrm{d}t} + r^2\frac{\mathrm{d}\omega}{\mathrm{d}t} = 0.$$

Then equation (6) simplifies to

$$\mathbf{a} = \left(\frac{\mathrm{d}^2 r}{\mathrm{d}t^2} - r\omega^2\right)\mathbf{n} \quad (8)$$

which is consistent with equation (1). [Board] It is important to realise here that equations (4), (5), (7) and (8) *do not depend explicitly on the existence of an inverse square force of attraction* and are therefore independent of Newton's Law of Gravitation and equation (1). These results follow whenever the *angular momentum of the particle P is constant*, so that P's orbit lies in a plane.

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4. <u>The role of Newton's gravitational force in orbital motion</u>

Returning now to Newton's postulate that there is gravitational force of attraction from the particle P towards the body B in the two-body scenario

and that equation (1) holds, it follows by comparing equation (8) with equation (1) and using equation (7), that

$$\frac{\mathrm{d}^2 r}{\mathrm{d}t^2} - \frac{L^2}{m^2 r^3} = -\frac{GM}{r^2} \qquad (9)$$

Once the differential equation (9) is solved for $r(t)$, $\theta(t)$ may be determined from equation (7), *i.e.*,

$$\frac{d\theta}{dt} = \frac{L}{mr^2}$$

so that particle P's trajectory about the body B may be found in the form $\mathbf{r}(t) = (r(t), \theta(t), 0)$. Using equations (9) and (7), we can show that *closed* orbits of P about B are possible and that these closed orbits are *ellipses*. I shall show this explicitly in a later lecture.

Notice that if the particle P has zero angular momentum, so that $L = 0$ in equation (9), then this equation reduces to

$$\frac{\mathrm{d}^2 r}{\mathrm{d}t^2} = -\frac{GM}{r^2}$$

which is the Newtonian equation for 'radial free-fall'. It follows from equation (7) that in this case, $\omega = 0$, so that θ is a constant. Hence, in this special case of P and B being both non-rotating, the particle P falls towards the body B along a path which is rectilinear in the common, relatively stationary, reference frame of the two objects. This is what we would expect, since it is only its angular momentum that keeps P in orbit about B.

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5. Conservation of angular momentum

We have seen that in the case of the two-body problem, the angular momentum of the orbiting particle P is conserved. Although the multi-body problem is notoriously intractable, in the sense of determining equations of orbits, according to Newtonian theory the total angular momentum of the system is conserved under certain conditions. Let us consider an *isolated* system of n particles, *i.e.*, not subjected to any external

influences. In order to constitute a system, the particles must interact in some way and in Newtonian theory all interactions are forces of some kind. It follows by *Newton's Third Law* (of action and reaction) that if all the forces are *central*, *i.e.*, act along the lines joining pairs of particles, then the *angular momentum of the system is always conserved*. This may be demonstrated as follows.

Let the angular momenta of the n particles be calculated relative to some fixed origin. Let the position vector of the ith particle relative to this origin be $\mathbf{r}_i$, its (linear) momentum $\mathbf{p}_i$ and its angular momentum $\mathbf{L}_i$. Let the total force exerted on the ith particle be $\mathbf{f}_i$ and let $\mathbf{f}_{ij}$ be the force exerted on the ith particle by the jth particle. It follows by Newton's Third Law that $\mathbf{f}_{ji} = -\mathbf{f}_{ij}$ for all i and j. Let the total angular momentum of the system be denoted by $\mathbf{L}$. Then by definition and equation (3),

$$\frac{\mathrm{d}\mathbf{L}}{\mathrm{d}t} = \sum_i \frac{\mathrm{d}\mathbf{L}_i}{\mathrm{d}t} = \sum_i \mathbf{r}_i \times \mathbf{f}_i = \sum_i \mathbf{r}_i \times \left(\sum_{j \neq i} \mathbf{f}_{ij} \right)$$

$$= \sum_i \sum_{j>i} (\mathbf{r}_i - \mathbf{r}_j) \times \mathbf{f}_{ij} = \mathbf{0}$$

since $\mathbf{f}_{ij}$ is parallel to $\mathbf{r}_i - \mathbf{r}_j$. [Note that in the double sum, $\mathbf{r}_i \times \mathbf{f}_{ij} = -\mathbf{r}_i \times \mathbf{f}_{ji} = -\mathbf{r}_j \times \mathbf{f}_{ij} \; \forall i \neq j$] Hence, $\mathbf{L}$ is a constant, as required.

It is this constancy of angular momentum in general which provides one of the foundations of the Pope-Osborne Angular Momentum Synthesis (POAMS), introduced in the next lecture.

In the next lecture, we begin to study POAMS at last and I will present the basic postulates of our synthesis and compare and contrast it with Newton's theory of gravitation.

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